Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 8 March 2026
Team ID:
Project Name: Quotes Recommendation Chatbot using NLP
Maximum Marks: 4 Marks

Technical Architecture

The technical architecture describes the **structure and interaction between different components of the chatbot system**.

In the **Quotes Recommendation Chatbot using NLP**, the system processes user queries, performs natural language processing to understand the user's intent, and retrieves relevant quotes from a dataset.

The architecture consists of the following components:

- User Interface (Web or Terminal)
- NLP Processing Module
- Quote Recommendation Engine
- Dataset Storage
- Response Generation Module

The system follows a **layered architecture** where user input flows through different processing layers before generating a response.

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interface where the user interacts with the chatbot	HTML, CSS, JavaScript / Terminal Interface
2	Application Logic-1	Handles user input processing and chatbot logic	Python
3	Application Logic-2	Performs Natural Language Processing operations such as tokenization and intent detection	NLTK / spaCy
4	Application Logic-3	Quote recommendation and response generation logic	Python
5	Database	Stores quotes categorized by topic	CSV Dataset / SQLite

S.No	Component	Description	Technology
6	Cloud Database	Optional cloud storage for quote dataset	Firebase / MongoDB Atlas
7	File Storage	Storage for quote datasets and configuration files	Local File System
8	External API-1	API used to integrate chatbot with web applications	Rasa REST API
9	External API-2	Optional API for advanced NLP models	HuggingFace NLP API
10	Machine Learning Model	Model used to understand user intent and recommend quotes	NLP Model (Intent Classification / TF-IDF)
11	Infrastructure (Server / Cloud)	Deployment environment for chatbot system	Local Server / AWS / Google Cloud

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Table-2 : Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source frameworks used for chatbot and NLP processing	Python, Rasa, NLTK
2	Security Implementations	Security measures to protect the system and data	Input validation, HTTPS, Authentication
3	Scalable Architecture	System can be extended by adding more datasets or advanced NLP models	Modular Architecture, Microservices
4	Availability	Chatbot should be accessible whenever the user sends a query	Web Server / Cloud Deployment
5	Performance	The chatbot should process user queries quickly and return quotes with minimal delay	Efficient NLP processing, caching techniques